

30066 – MACROECONOMICS

(8 CFU)

CLASS : BIEM 14 & 19
2024 – 25

Lecture 18 (Ch. 15)

Financial Markets and Expectations: the
stock market

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WHERE WE ARE

Fundamentals

1. Macroeconomic variables
 - Elements of national accounting
2. The short run
 1. Goods markets
 2. Financial markets
 1. Without commercial banks
 2. With commercial banks
 3. IS-LM model
3. The medium run
 1. Labor market
 2. Phillips curve
 3. IS-LM-PC model
 1. Medium and short run equilibrium
 2. Demand and supply shock

Extensions

4. Expectations, financial markets, and economic policies
 1. Financial markets and expectations
 2. Consumption, investment, and expectations
 3. Output, economic policies, and expectations
5. Open economy
 1. Exchange rates, trade, and international returns
 2. Goods markets in an open economy
 3. Financial markets and economic policies in an open economy
6. Public debt and fiscal policy
7. From the Great Recession to Quantitative Tightening
 - From a housing crisis to a financial crisis
 - Unconventional monetary policies

S&P 500

5.776,65

↑ 671,56%

+5.027,95 MAX

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1G

5G

1M

6M

YTD

1A

5A

MAX



STOCK MARKET, STOCK PRICES AND EXPECTATIONS

How do firms finance their activities?

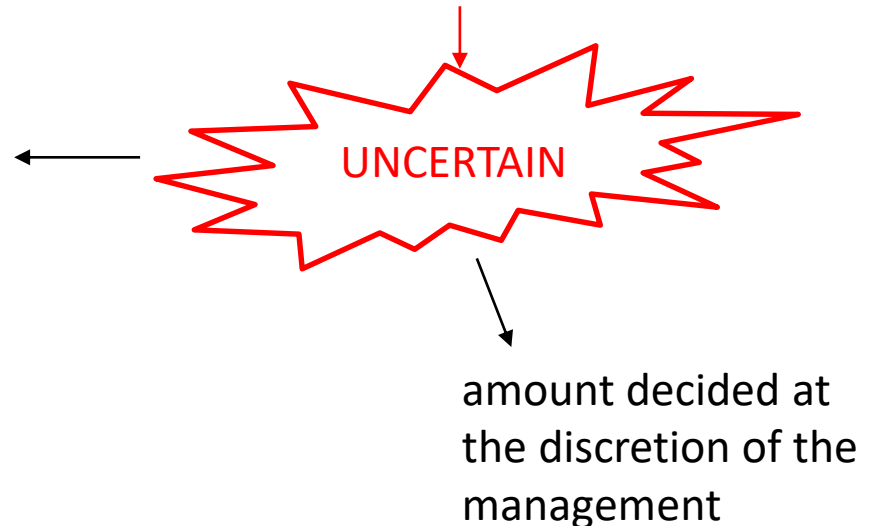
1. Internal financing: use their profits
2. External financing: through debt
 - Banks' loans
 - Corporate bonds

3. Equity: issuing stocks/shares

Investors
participate to the firm ownership
and could obtain part of the profits
as DIVIDENDS

Depend on profits that depends
on macro economic dynamic:

if GDP ↑ , ↑ Income, ↑ firms
sales => ↑ profits => ↑
dividends



United States Stock Market Index (US500)



source: tradingeconomics.com

Italy Stock Market Index (IT40)



source: tradingeconomics.com

What causes the big fluctuations of the market? The major booms and crashes of the stock market?

→ Macro features that influence the demand of stocks and their price

- Stocks can be more or less attractive to investors
- Their attractiveness determines how much investors are willing to pay
- Based on D and S, the stock market determines the price



when relative $D \uparrow$ their price $\uparrow$

What factors influence the relative D for stocks?

1. Firms profitability: firms ability to generate profits and distribute dividends

If at time t I buy a stock, I expect to receive a stream of dividends starting from $t+1$ until the last period I hold the stock in my portfolio.

$$(\mathbb{E}D_t, D_{t+1}^e, \mathbb{E}D_{t+3}^e, \dots)$$

If $D^e \uparrow$, stocks are more attractive $\Rightarrow$ their demand $\uparrow \Rightarrow$ their price $\uparrow$

2. Current and expected short term interest rates: they determine attractiveness of stocks wrt bonds

If $i/i^e \downarrow$, bonds are less attractive for investors than stocks $\Rightarrow$ demand for stocks $\uparrow \Rightarrow$ their price $\uparrow$

3. Attitudes towards risk

If agents become less risk averse they tend to demand more stocks $\Rightarrow$ their price $\uparrow$

They depends on macroeconomic factors (phases of business cycles, MP, FP, geopolitical instability, trust in institutions ...)

- Assume you want to invest 1€ for 1 year
- 2 options: a 1-year bond and a stock

i_{1t}

$€Q_t$ = *Ex-dividend price* of a stock at time t , after the payment of current period dividend => the next dividend will be at $t+1$

t

$t+1$

Amount invested

Returns from holding the asset for one year

1 YEAR BOND

1€



$(1 + i_{1t})€$

Certain return

STOCK

1€



$\frac{1}{€Q_t}(€D_{t+1}^e + €Q_{t+1}^e)$

Expected return

For each stock held at time t , at $t+1$:

- 1) You receive the dividend $€D_{t+1}^e$
- 2) You can resell the asset on the stock market at the expected price prevailing on the stock market next year $€Q_{t+1}^e$

Stocks are riskier assets wrt bonds:

both dividends and future prices are unknown at time t

=> investors require compensation for the higher risk → equity premium, x

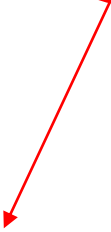
The 1 year expected return of the stock must be greater than the return of a bond in order to trade both assets (arbitrage condition)

$$\underbrace{\frac{\mathbb{E}D_{t+1}^e + \mathbb{E}Q_{t+1}^e}{\mathbb{E}Q_t}}_{\text{1 year expected return of a stock}} = \underbrace{1 + i_{1t}}_{\text{Return of a risk-free bond}} + \underbrace{x}_{\text{Equity Risk premium}}$$

Rearranging:

$$\mathbb{E}Q_t = \underbrace{\frac{\mathbb{E}D_{t+1}^e}{1 + i_{1t} + x}}_{\text{PDV of dividends expected for the next period}} + \underbrace{\frac{\mathbb{E}Q_{t+1}^e}{1 + i_{1t} + x}}_{\text{PDV of the price at which the stock can be sold next year}} \rightarrow \text{Price is the expected PDV of future payments}$$

Repeating the intuition for the next year and the subsequent one, assuming n high enough, we can write:

$$\textcircled{\text{€}Q_t} = \frac{\text{€}D_{t+1}^e}{1 + i_{1t} + x} + \underbrace{\frac{\text{€}D_{t+2}^e}{(1 + i_{1t} + x)(1 + i_{1t+1}^e + x)} + \dots}_{\text{PDV of dividends expected from } t+2 \text{ onward, up to the last period the stock is held in the portfolio}}$$


The stock price is the expected present discounted value of future expected dividends:
it is the fundamental value of the stock
⇔ it reflects the underlying fundamentals of the economy

$$\text{€}Q_t = f(\text{€}D_{t+1}^e, i_{1t}, i_{1t+1}^e, i_{1t+2}^e, \dots, x)$$

Current price of stocks increases if the relative demand of stocks increases

❖ Higher expected future dividends lead to a higher real stock price:

If expected GDP $\uparrow \Rightarrow \text{€}D^e \uparrow \Rightarrow$ stocks more attractive for investors $\Rightarrow \uparrow$ demand for stocks $\Rightarrow \text{€}Q_t \uparrow$

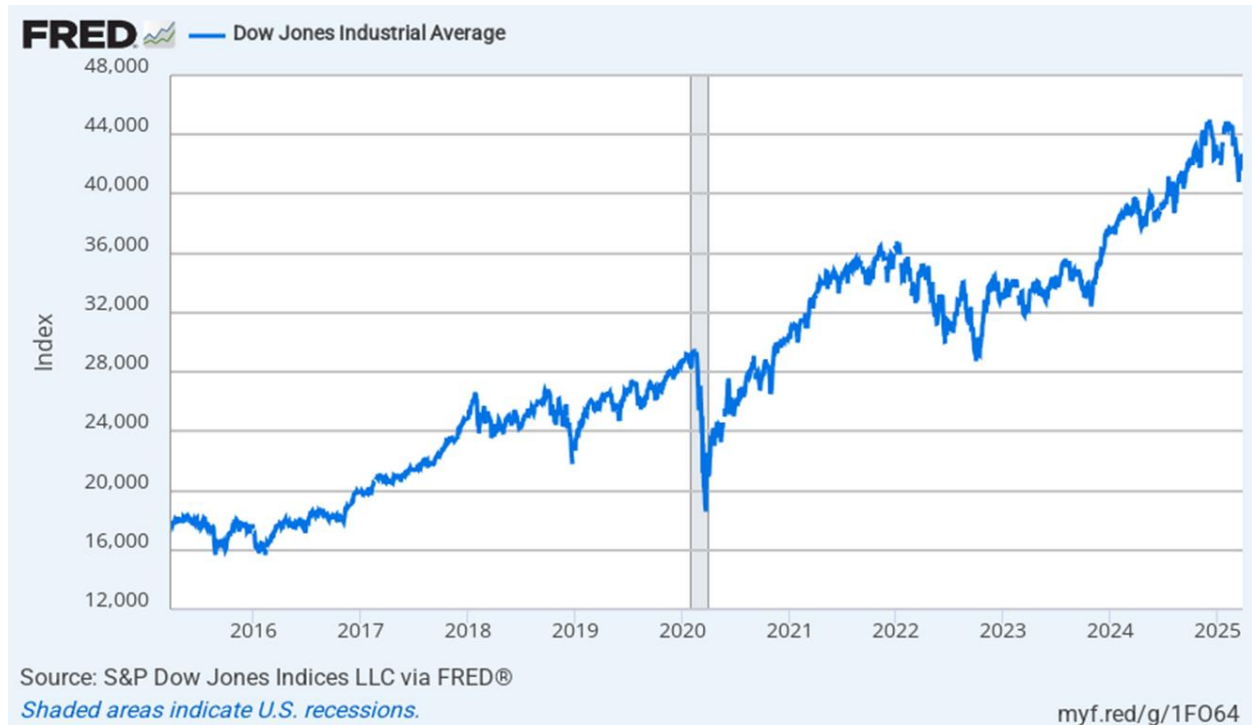
❖ Lower current and expected future one-year real interest rates lead to a higher real stock price:

If $i \downarrow$ and/or $i^e \downarrow \Rightarrow$ bonds are less appealing for investors that ask for more stocks and less bonds
 $\Rightarrow \text{€}Q_t \uparrow$

Or symmetrically, given the expected dividends their PDV increases, to hold a stock becomes worthwhile for investors, they ask for more stocks $\Rightarrow \Rightarrow \text{€}Q_t \uparrow$

❖ A lower risk premium leads to a higher stock price

If $x \downarrow \Rightarrow$ agents less risk averse $\Rightarrow$ ask for more stocks $\Rightarrow \text{€}Q_t \uparrow$



THE STOCK MARKET AND THE ECONOMY

Can stock market fluctuations be forecasted?

Not most of them! Major movements in stock prices are unpredictable

The stock market follows a random walk with some regularities

What are the possible impacts of economic shocks/policies on the stock market?

It depends on whether

1) Shocks are EXPECTED or UNEXPECTED



Already incorporated in
the current value of stocks
⇒ the fundamental value
does not change
⇒ the stock market does
not react
⇒ $\Delta \epsilon Q_t = 0$



Not incorporated in the current value of
stocks
⇒ the fundamentals behind the stock
change in an unexpected way
⇒ the stock market reacts
⇒ $\Delta \epsilon Q_t \neq 0$

2) Announced changes in the economy are CREDIBLE or NOT CREDIBLE



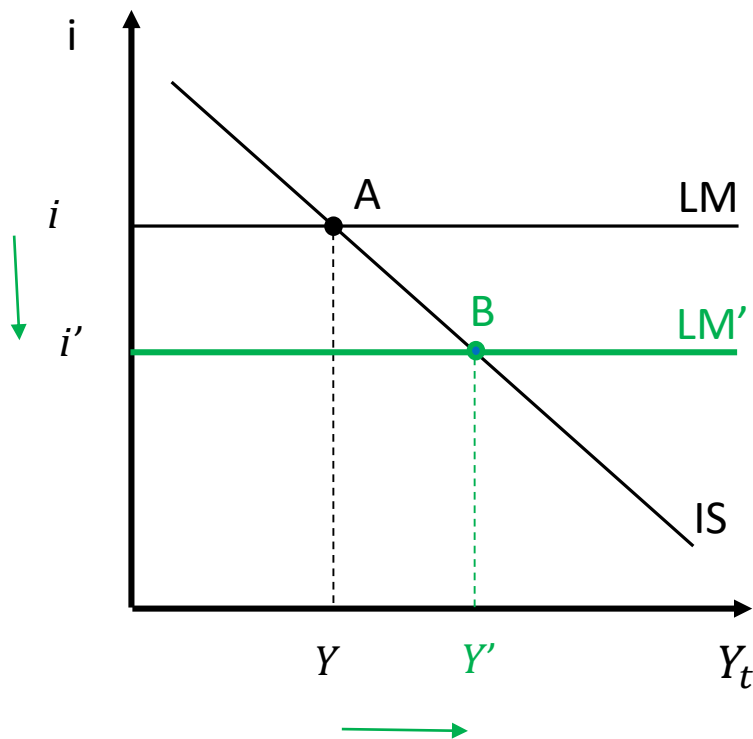
the stock market reacts



the stock market does not
react

1) The CB implements an unexpected expansionary MP in period t , no changes are expected for $t+1$

The economy today: period t



In t ,

- the short term interest rate ↓: given future expected dividends, their present value ↑
 $\Rightarrow$ stocks more attractive for investors
 $\Rightarrow$ their demand ↑
 $\Rightarrow$ their current price ↑
- output ↑ :
 $\Rightarrow$ Firms expected profits ↑
 $\Rightarrow$ Stream of expected dividends ↑
 $\Rightarrow$ stocks more attractive for investors
 $\Rightarrow$ their demand ↑
 $\Rightarrow$ their current price ↑

In $t+1$ no changes are expected $\Rightarrow$ no effect on the stock market

Overall ,

$$\Delta \epsilon Q_t > 0$$

$\Rightarrow$ the fundamentals behind the stock change in an unexpected way and the stock market reacts

2) The economy is in a strong recession and in period t the CB implements an expansionary MP already announced

In t , as expected the short term interest rate $\downarrow$, investment in physical capital $\uparrow$, the aggregate demand increases and output $\uparrow$

Since these changes were already expected by agents the stock market does not react

$\Rightarrow$ No changes in the fundamentals behind the stock value

$\Rightarrow$ No changes in the price of stocks

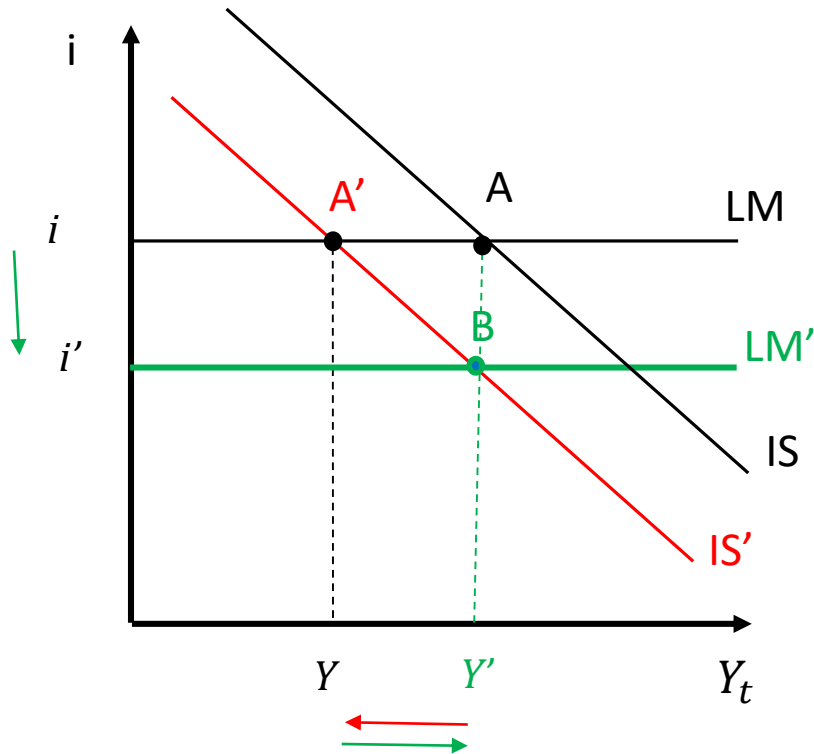
$$\Rightarrow \Delta \epsilon Q_t = 0$$

TRUE OR FALSE

An unexpected default of a major financial institution definitely leads to a reduction in stock prices

False

Unexpected default of a big bank and the CB intervenes to avoid changes in output



- Uncertainty in the financial markets increases the borrowing rate, $\downarrow I$, $\downarrow Z$ and $\downarrow Y$ (IS to the left)
- The CB intervenes to keep the GDP constant and to avoid a recession $\Rightarrow$ expansionary MP, the CB buys bonds and $i \downarrow$ (LM down)

In t , the short term interest rate $\downarrow$ and given future expected dividends, their present value $\uparrow$

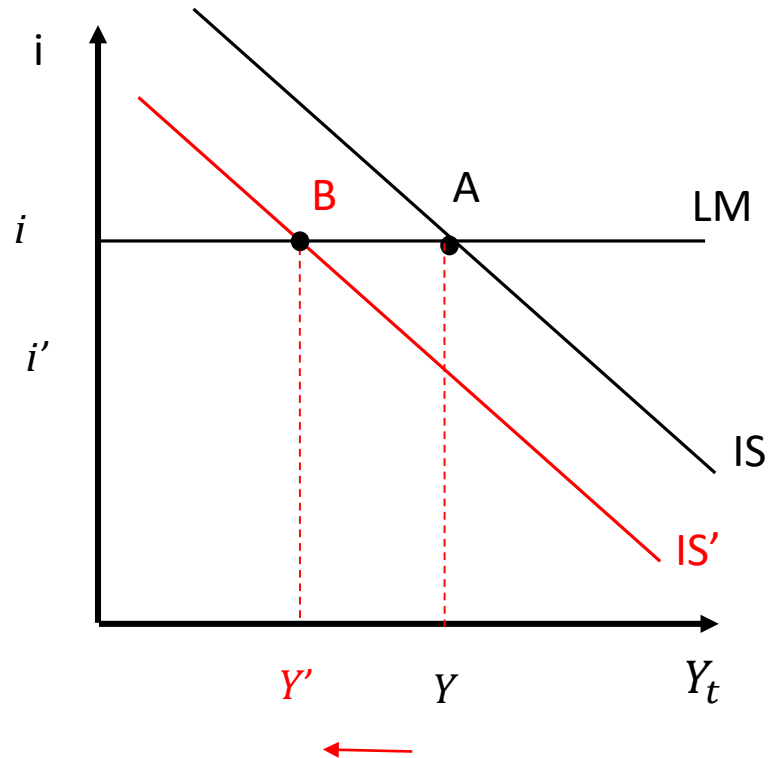
$\Rightarrow$ stocks more attractive for investors

$\Rightarrow$ their demand $\uparrow$

$\Rightarrow$ their current price $\uparrow$

$\text{€}Q_t \uparrow$

Unexpected default of a big bank and the CB does not intervene



- Uncertainty in the financial markets increases the borrowing rate, $\downarrow I$, $\downarrow Z$ and $\downarrow Y$ (IS to the left)

In t in an unexpected way output $\downarrow$:

- $\Rightarrow$ Firms expected profits $\downarrow$
- $\Rightarrow$ Stream of expected dividends $\downarrow$
- $\Rightarrow$ stocks less attractive for investors
- $\Rightarrow$ their demand $\downarrow$
- $\Rightarrow$ their current price $\downarrow$
- $\Rightarrow \epsilon Q_t \downarrow$

TRUE OR FALSE

In period t , the central bank announces a monetary contraction to be implemented in $t+1$. Assuming the announcement has no impact on today's economy and was not expected by financial markets, stock prices would decrease.

False

The effect of the announcement depends on whether the Central Bank
credible or not credible

Not credible

- Agents believe that the policy will not be implemented
- No changes in expectations
- No changes in fundamentals

$$\textcolor{red}{\epsilon \Delta Q_t =}$$

Credible

- Change in expectations and in expected fundamentals

$$Y^e \downarrow \quad i^e \uparrow$$

$$\textcolor{red}{\epsilon Q_t \downarrow}$$

Stock prices change, even if **nothing** happens **today**

QUICK CHECK

If in the economy there is an expected wave of optimism, the stock market reacts _____ if the Central Bank has a target in terms of _____?

A. more, interest rate

B. less, interest rate

C. more, money supply

B. less, reserve requirement

QUICK CHECK

What happens if the Central Bank announces a credible monetary contraction for the next period?

A. Stock prices increase immediately

B. Stock prices decrease today

C. There is no effect until the policy is implemented

D. The effect depends on the announcement being expected

QUICK CHECK

If a shock is already expected and priced in by the market, what is the likely effect on stock prices when it happens?

- A. Stock prices drop sharply
- B. Stock prices rise suddenly
- C. There is no significant change**
- D. Investors sell off their stocks

QUICK CHECK

The impact of an economic policy announcement depends mostly on:

A. Whether the Central Bank has political support

B. If markets believe the policy will be implemented

C. How high interest rates are

D. The level of public debt

TAKE AWAY AND KEY CONCEPTS

- Stock prices as expected present discounted value of future payments
- Stock prices are influenced by:
 - Current economic conditions
 - Expected future economic developments
- Reaction of stock prices depends on:
 - Expectations
 - Source of the shock
 - Credibility of economic policy authorities